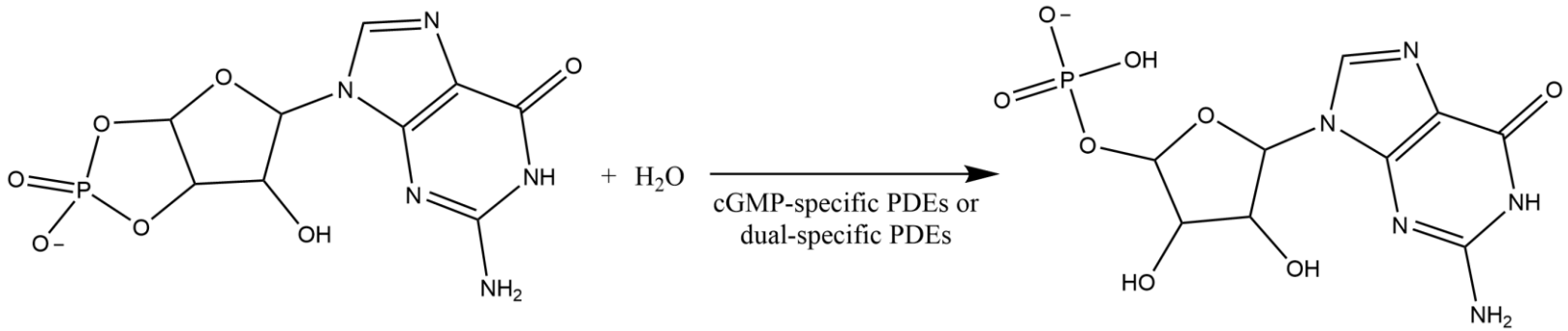
IMPORTANT NOTES

1. **REVIEW THE LITERATURE ABOUT THE PEOTEIN CAREFULLY BEFORE YOU START TO WORK ON IT!!!!!**
2. **Make sure you know what you are doing!**
3. Ask if you have any questions on the reason behind a certain step, on the theory, or when you face any technical issues!
4. **WARNING: CHARMM-GUI may have issues when parameterizing charged ligands.**
5. To make it simple, choose a small enzyme with a neutral **substrate**
   - **You need to use the ES-complex to investigate the effect of metal on catalytic activity**

Introduction background of PDE9A

- **cGMP-specific Phosphodiesterase:** Hydrolyses second messenger cGMP



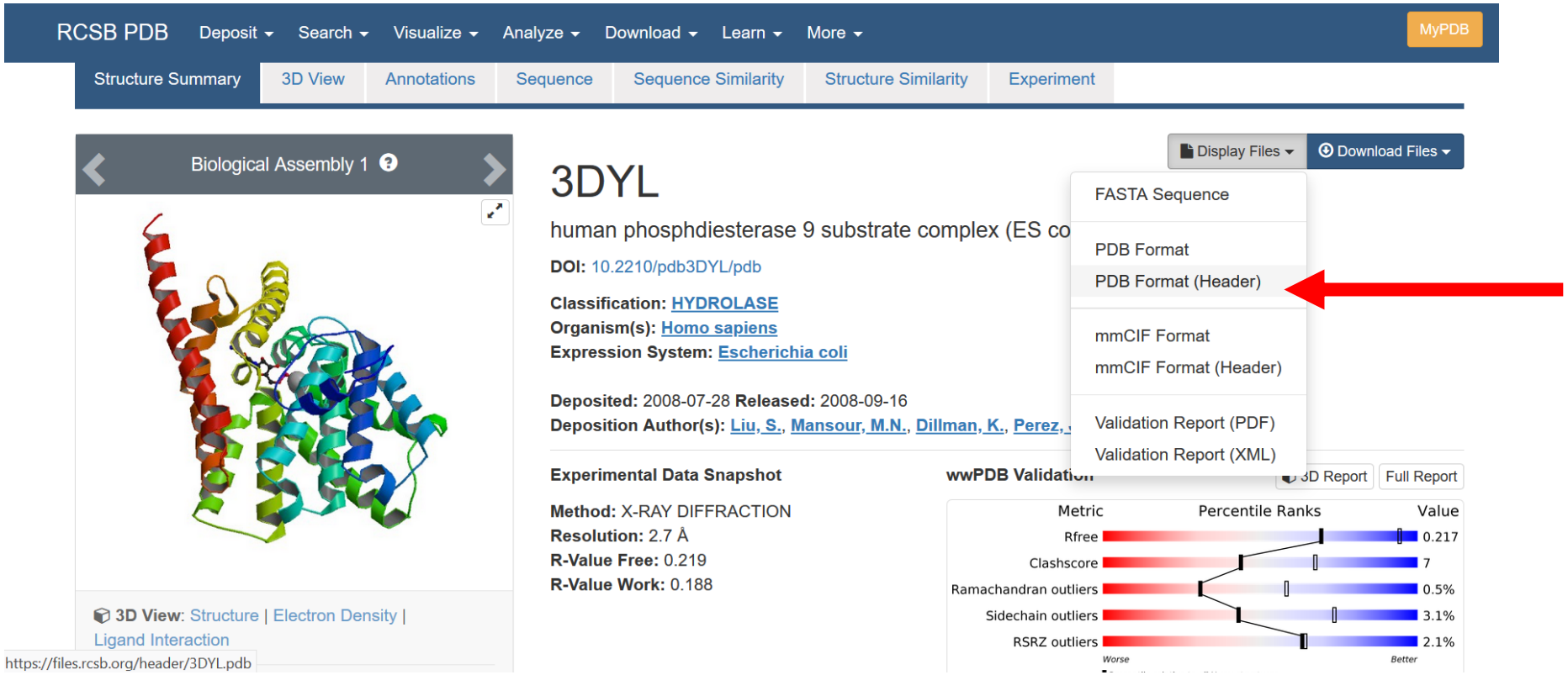
- **Potential drug target of sickle cell disease …**
- **2 metal ions that plays important roles in the catalytic process**

0. Literature Review Useful information from literature

- **Catalytic pocket residues protonation**
  - **HIS252:** protonated at both delta and epsilon position (forgot which paper)
  - **Others:** guess from hydrogen bonds using UCSF Chimera
- **Substrate cGMP**
- **Crystal water**
  - **The water molecule between two metal ions is treated as OH^-**
    - ✓ (ref doi: 10.1529/biophysj.106.086835 10.1021/ja005529a)
  - All the water molecules are kept (Important to keep the water coordinating with metal!)
- **Metal ions and activity of enzyme** (ref doi: 10.1073/pnas.0708850105)
  - **Inactive:** Zn^{2+}-Zn^{2+}, Na^{+}-Na^{+}
  - **Active:** Zn^{2+}-Mg^{2+}, Mn^{2+}-Mg^{2+}, Mn^{2+}-Mn^{2+}, Mg^{2+}-Mg^{2+}

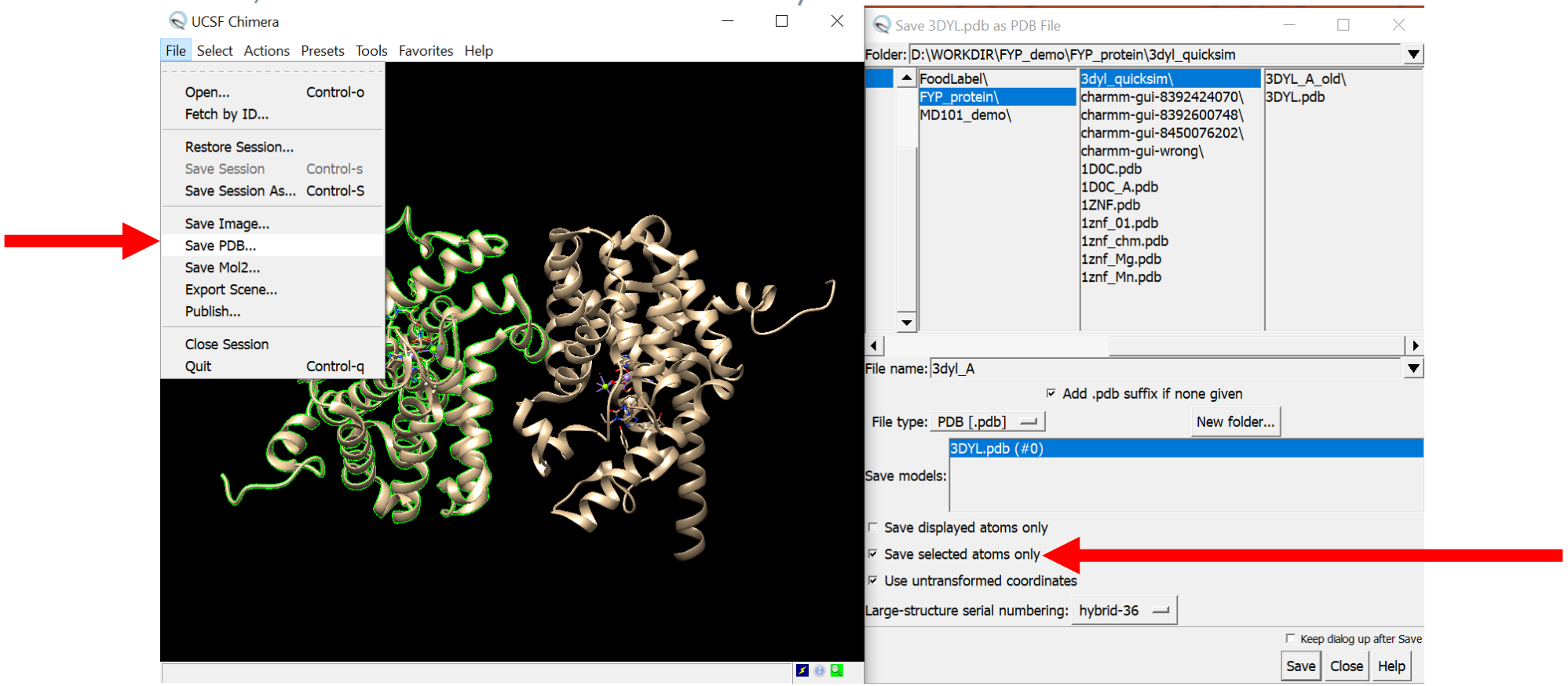
1a. Check the PDB file

- Search 3DYL in https://www.rcsb.org
- Click Display Files > PDB Format (Header). Check the following items:
  - **Is it a mutated structure?** Search MUTA to look for this.
  - **Any non-terminal missing residues?** Search MISS to look for this.



1b. Download the Crystal Structure and Extract Interested Part

- If you are happy with this PDB file, download the file from the RCSB PDB
- Open the PDB file with UCSF Chimera
  - Select > Chain > A
  - File > Save PDB, select “save selected atoms only”. Give it a name and save.

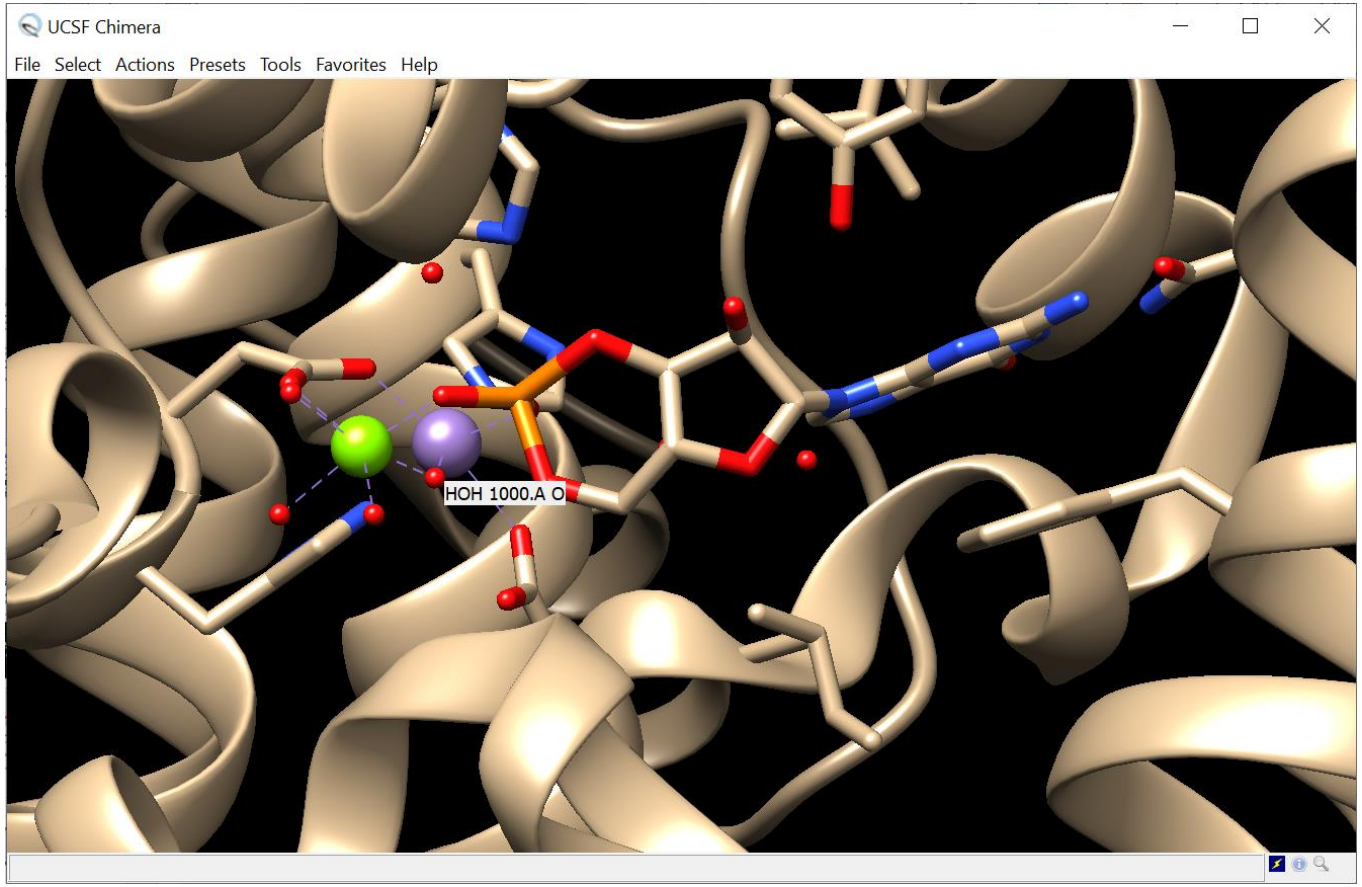


1c. Modify the PDB file

- Remember what's in the literature?

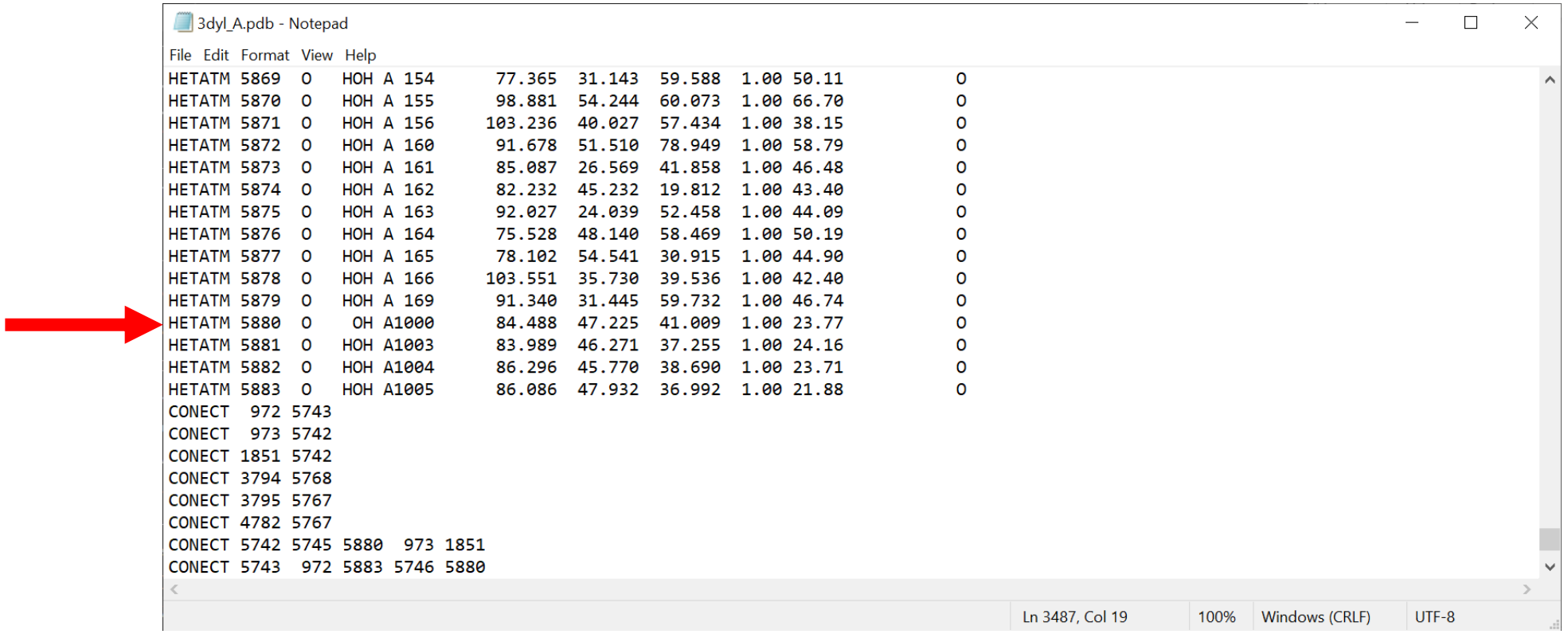
  "The water molecule between two metal ions is treated as OH^- "

- Find this water in chain A and mark its residue ID
  - Its HOH 1000



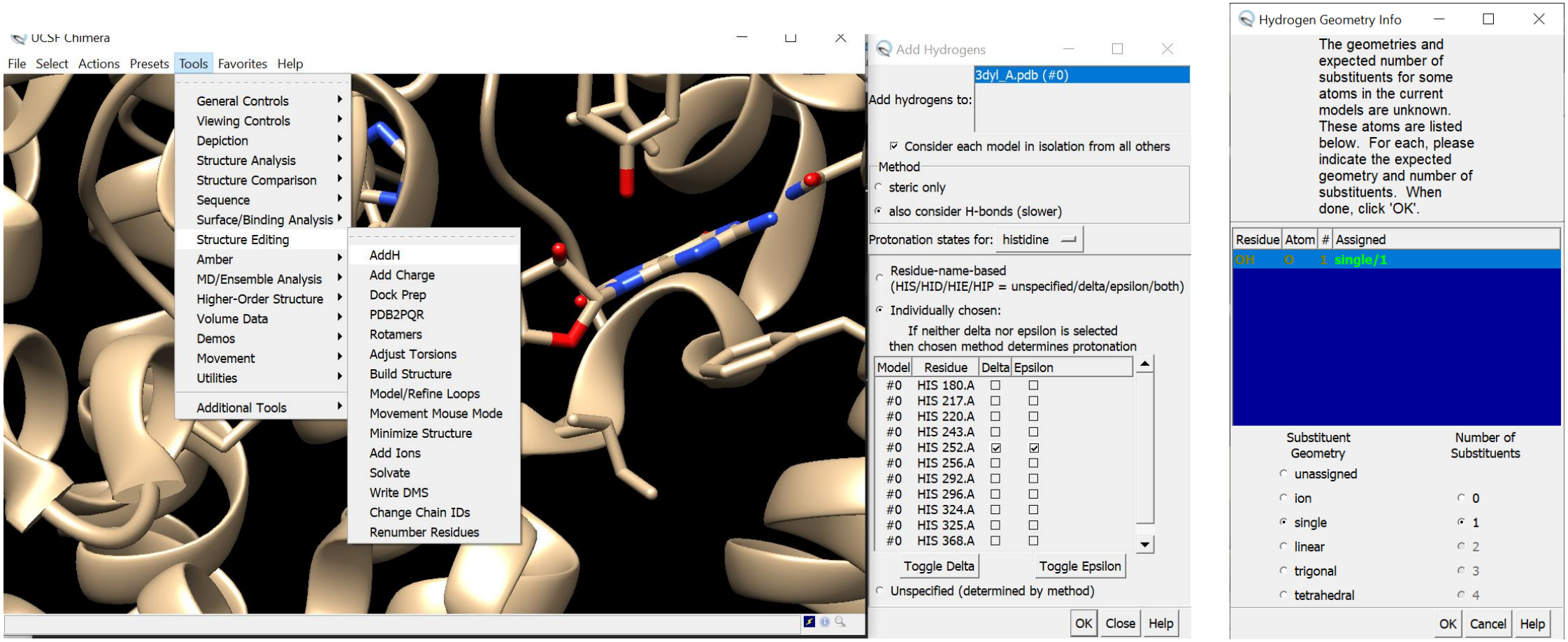
1c. Modify the PDB file

- Open 3dyl_A with Notepad
- Rename “HETATM 5880 O   HOH A1000” to “HETATM 5880 O   OH A1000”
- Save the modified PDB



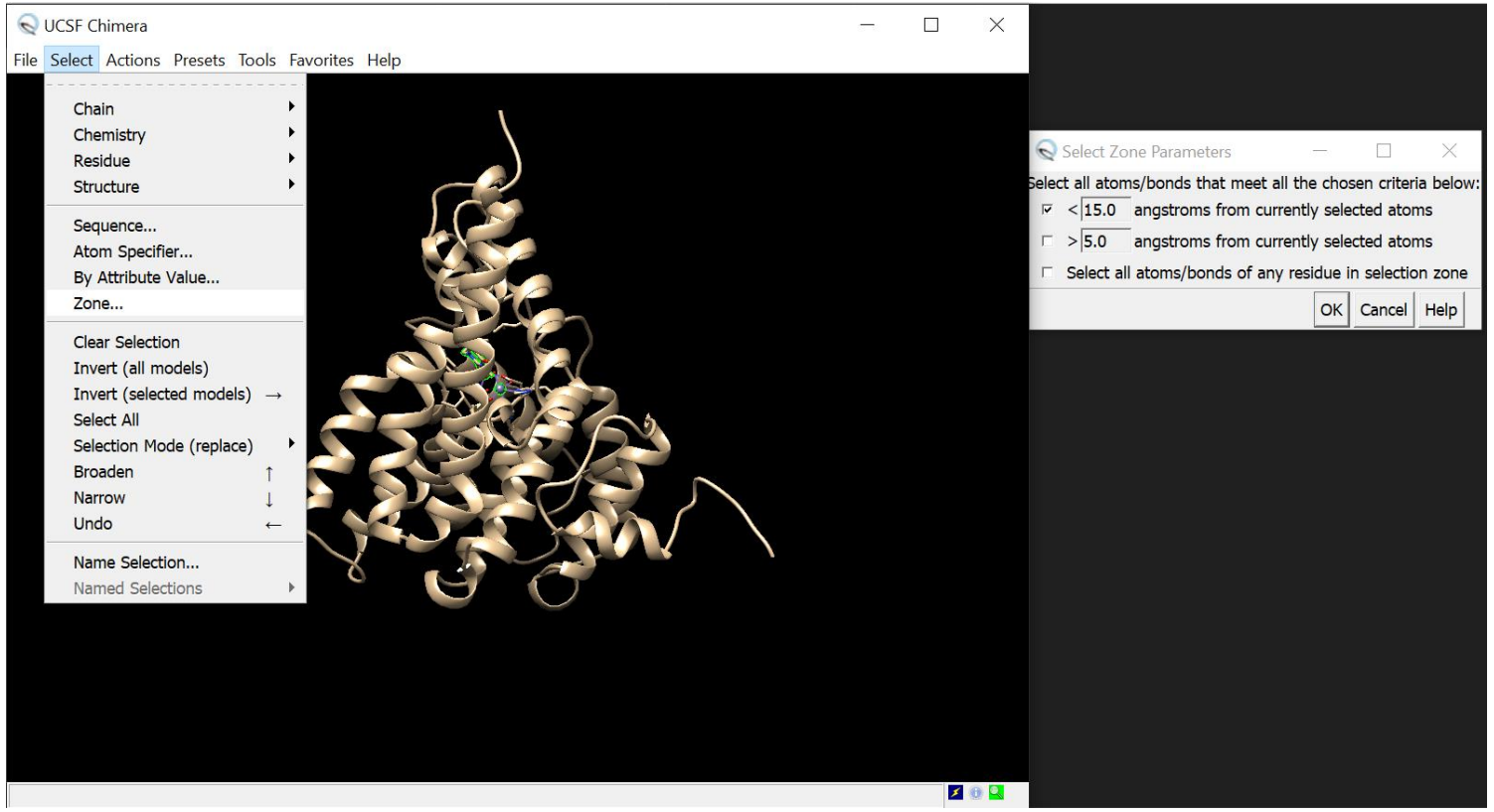
1c. Add H

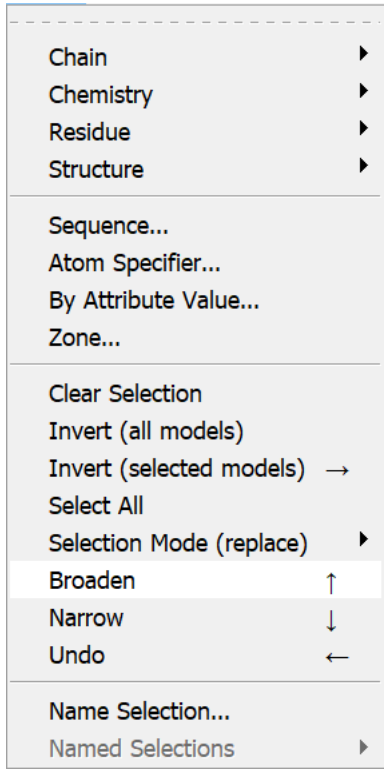
- Open the modified PDB in chimera



1d. Select residues within 15A of metal and ligand

- **Select the metal ions and the substrate**
- Select > Zone, type 15.0 in the first box, click ok
- Select > broaden to select the whole residues





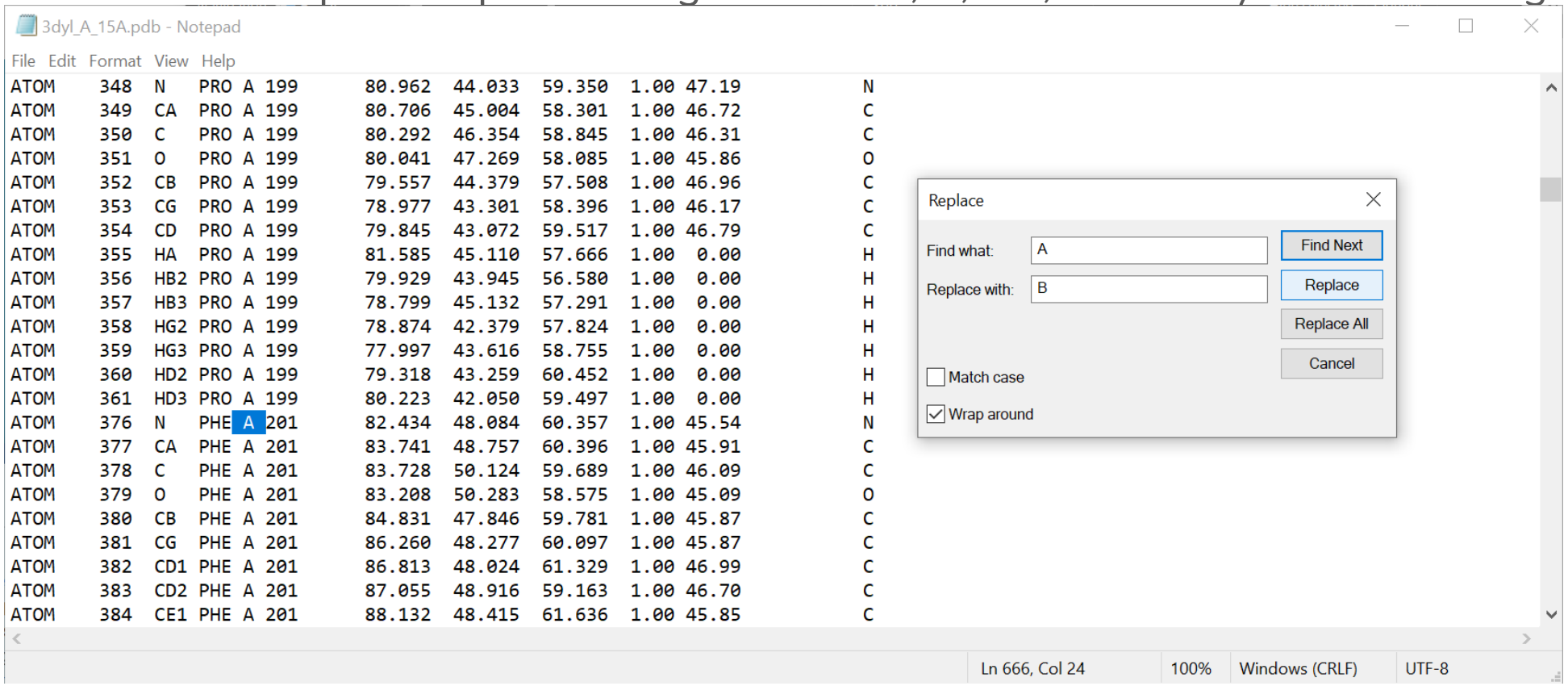
- Save the selected atoms like before

1d. Check and modify the trimmed structure

- Open the saved pdb with chimera. Inspect it carefully
- Remove or add back the residues of small protein segments as you wish
- Save the PDB file

1d. Check and modify the trimmed structure

- Open the pdb with Notepad
- Rename chain ID the separated protein fragments to B, C, D..., if not they'd be linked together



- You may also need to rename HIS to HSD, HSE or HIP based on histidine protonation. Use a notepad to do so.

2a. Upload to CHARMM-GUI

- Hold on!!!
- In this case: CHARMM-GUI failed to generate the topology of negatively charged PCG and OH^- so I need to use other methods
- Let me know if you face this problem. I use 3DYL as an example just to illustrate the importance of literature review and the usual method of doing a real research.
- From now on I'll demonstrate what may happen if you blindly use a simple method like a "black box", without knowing the background info or have a clear idea of what you are doing.
  - You may get a "reasonable" simulation results without any errors, but it is **not realistic at all**!
- Now I'll switch to 2HD1, the enzyme-**inhibitor** complex (usually for drug design) of the same enzyme that have the OH- missing, without any modification to pass all the problematic steps of charged ligands parameterization in CHARMM-GUI. I have repeated all the steps in 1 using 2HD1 chain A.
- **THIS IS NOT THE CORRECT WAY TO SOLVE THIS ISSUE, JUST FOR A QUICK DEMONSTRATION OF THE STEPS TO HAVE A QUICK ESTIMATION OF THE INTERACTION ENERGY BETWEEN THE PROTEIN AND THE SUBSTRATE.**

2a. Upload to CHARMM-GUI

- http://www.charmm-gui.org/?doc=input/solution
- Select ALL the residues that you want to keep
- Use CSML search for ions
- For the ligand: default settings may work, but not always.
- You may want to change the terminal patching of the segments.
- **In this trial: no position restraints applied for the fragments in production run (because it is too tedious... Im lazy)**

3a. Modify the input files generated by CHARMM-GUI

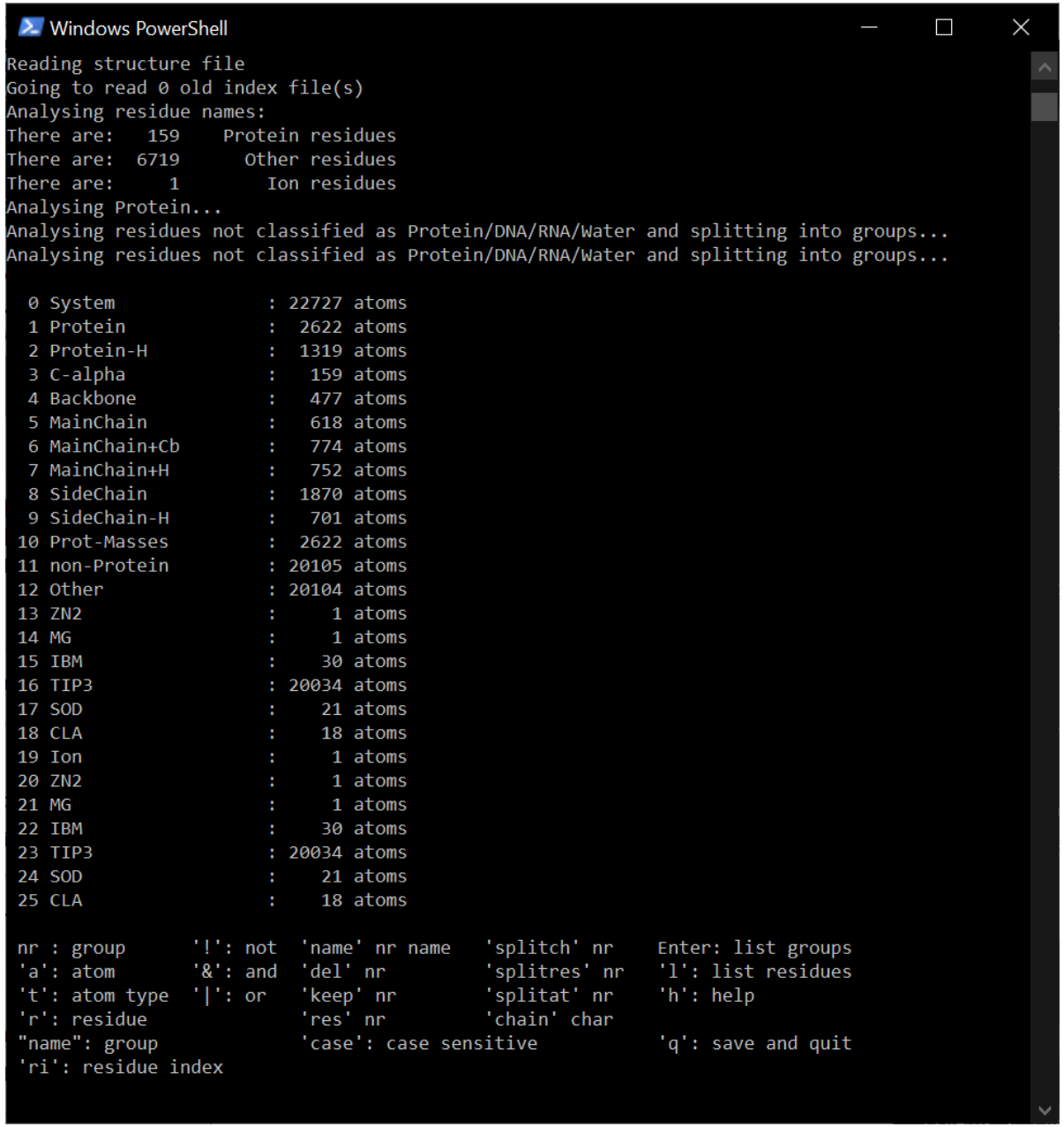
- **Modify the mdp files to be run with the simulation_template.cmd**
  - Similar to the 1ZNF simulation: add a restrained NPT step4.2 before production run
  - For step5, use 500000 steps (=1ns) first as a timing test.
  - **Use a smaller number (e.g. remove a zero) for nst???out to take more data in step5_production.mdp**

3bi. Generate index groups

- Open the gromacs folder after extracting the files from CHARMM-GUI
- Open the folder with PowerShell or cmd, execute:

```
gmx.exe make_ndx -f .\step3_charmm2gmx.pdb -o nrg_grp.ndx
```

  - Check the list of index groups, then follow a similar procedures as the next screenshots
  - When you work on other enzymes, IBM would be the name of your ligand instead.

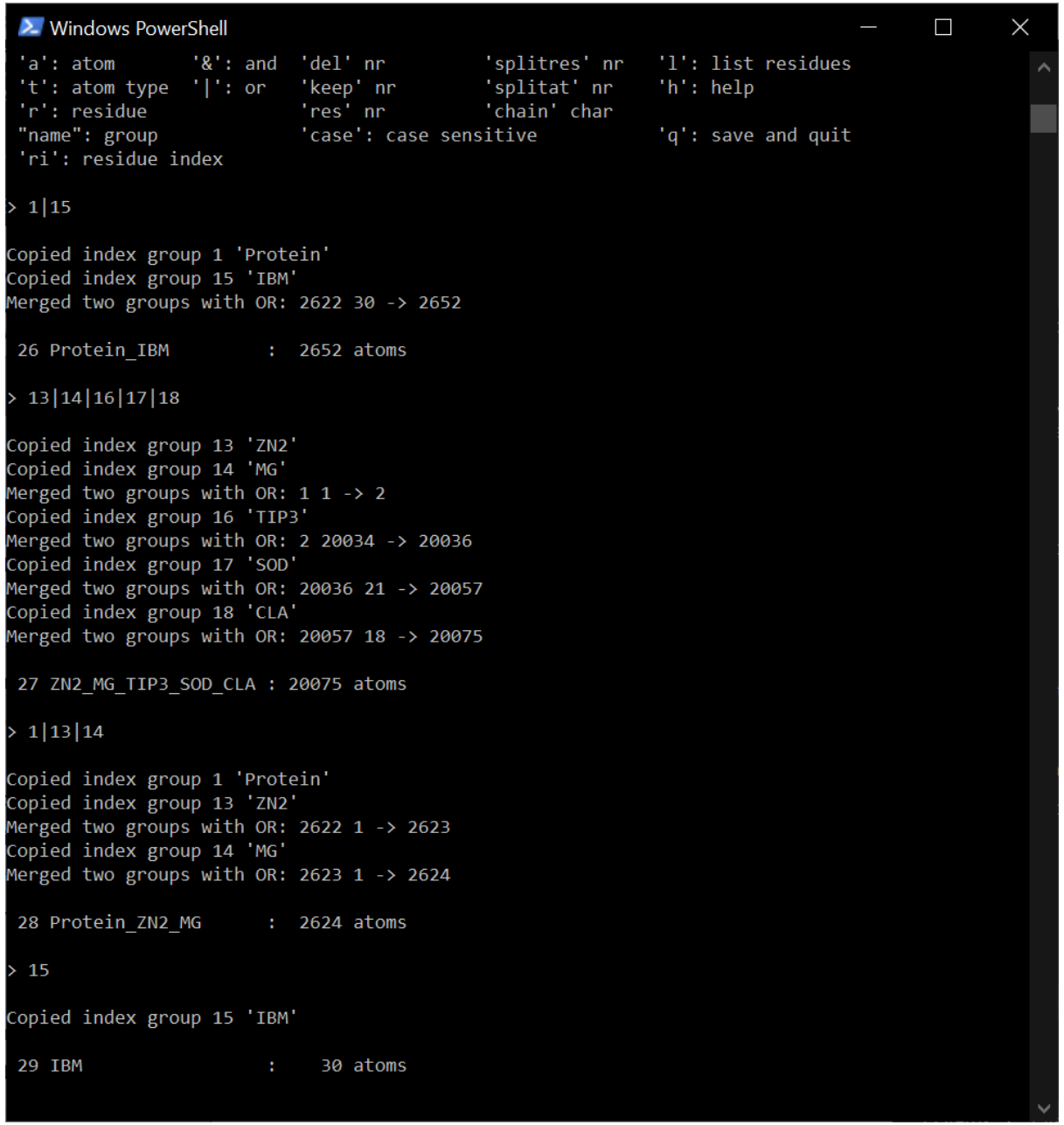


3bii. Generate index groups

- Open the gromacs folder after extracting the files from CHARMM-GUI
- Open the folder with PowerShell or cmd, execute:

gmx.exe make_ndx -f .\step3_charmm2gmx.pdb -o nrg_grp.ndx

  - When you work on other enzymes, IBM would be the name of your ligand instead.

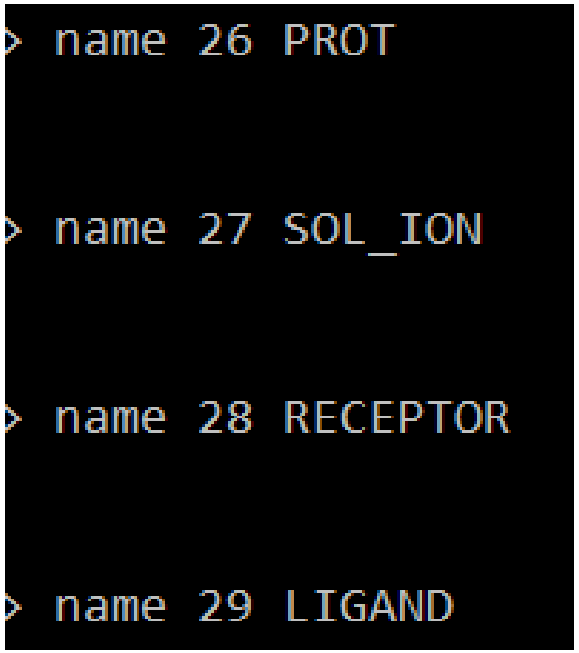


3biii. Generate index groups

- Open the gromacs folder after extracting the files from CHARMM-GUI
- Open the folder with PowerShell or cmd, execute:

```
gmx.exe make_ndx -f .\step3_charmm2gmx.pdb -o nrg_grp.ndx
```

  - When you work on other enzymes, IBM would be the name of your ligand instead.

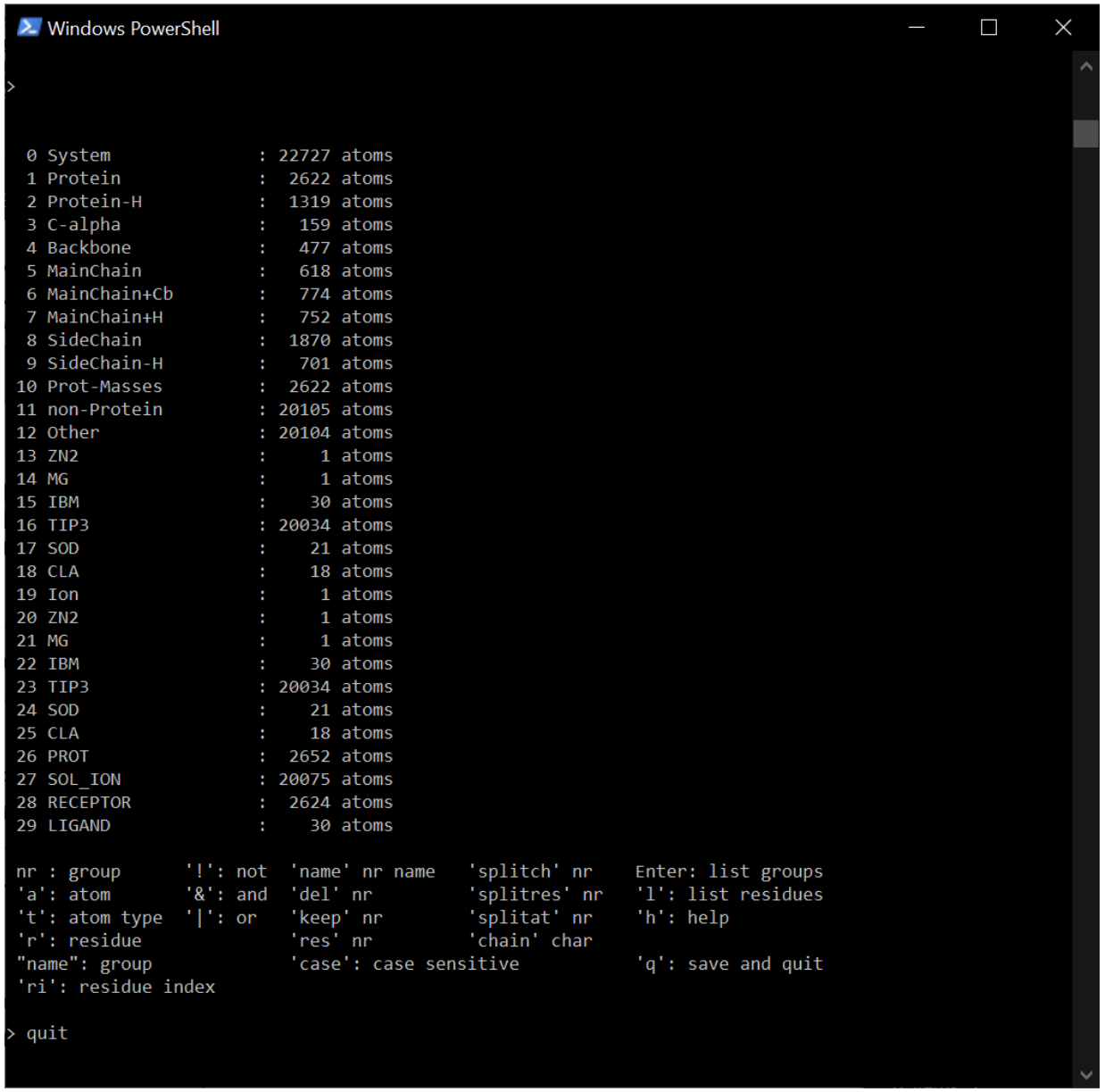


3bvi. Generate index groups

- Open the gromacs folder after extracting the files from CHARMM-GUI
- Open the folder with PowerShell or cmd, execute:

```
gmx.exe make_ndx -f .\step3_charmm2gmx.pdb -o nrg_grp.ndx
```

  - When you work on other enzymes, IBM would be the name of your ligand instead.
  - After renaming the new groups, press enter to check the new list of index groups
  - If it is ok, type quit



3c. Add a mdp file with energy groups for interaction energy calculation

- Copy step5_production.mdp to step6_ie.mdp
- Add the following line to step6_ie.mdp:

```
energygrps                  = RECEPTOR LIGAND
```

3d. Appling position restraint (im lazy so I skipped this ...)

- **If you would like to apply position restraints at the terminal of each segments:**
  - Too much to type in this step...
  - Extract all the chains independently from step3 pdb file, to e.g. chainA.pdb chainB.pdb ...
  - Use gmx.exe make_ndx to make a group for the terminal of each chain, and save e.g. chainA.ndx chainB.ndx ...
  - then use e.g. gmx.exe genrestr –f chainA.pdb –n chainA.ndx –o posre_terA.itp ... repeat for B, C, D...
  - Then, in your topol.top, right after the line #include "toppar/PROA.itp“, add
    #ifdef POSRE_TER
    #include posre_terA.itp
    #endif
    - Repeat for B, C, D ...
  - Then, at the top of your step5_production.mdp and step6_ie.mdp, add:
    define = -DPOSRE_TER
  - In this case I need to repeat the same thing from A to R ...
  - Since it is too tedious im skipping it here... but beware, without restraints the small segments may move away ...
  - Actually you can write a script to do it automatically... but im lzay

3e. Copy simulation_template

- Similar to the 1znf simulation: Copy the simulation_template.cmd with the correct path to gmx.exe to this folder
- **Add the following lines to this simulation_template.cmd:**

D:\WORKDIR\FYP_demo\FYP_protein\2hd1_wrong\charmm-gui-8641679700\gromacs\simulation_template.cmd - Notepad++

File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?

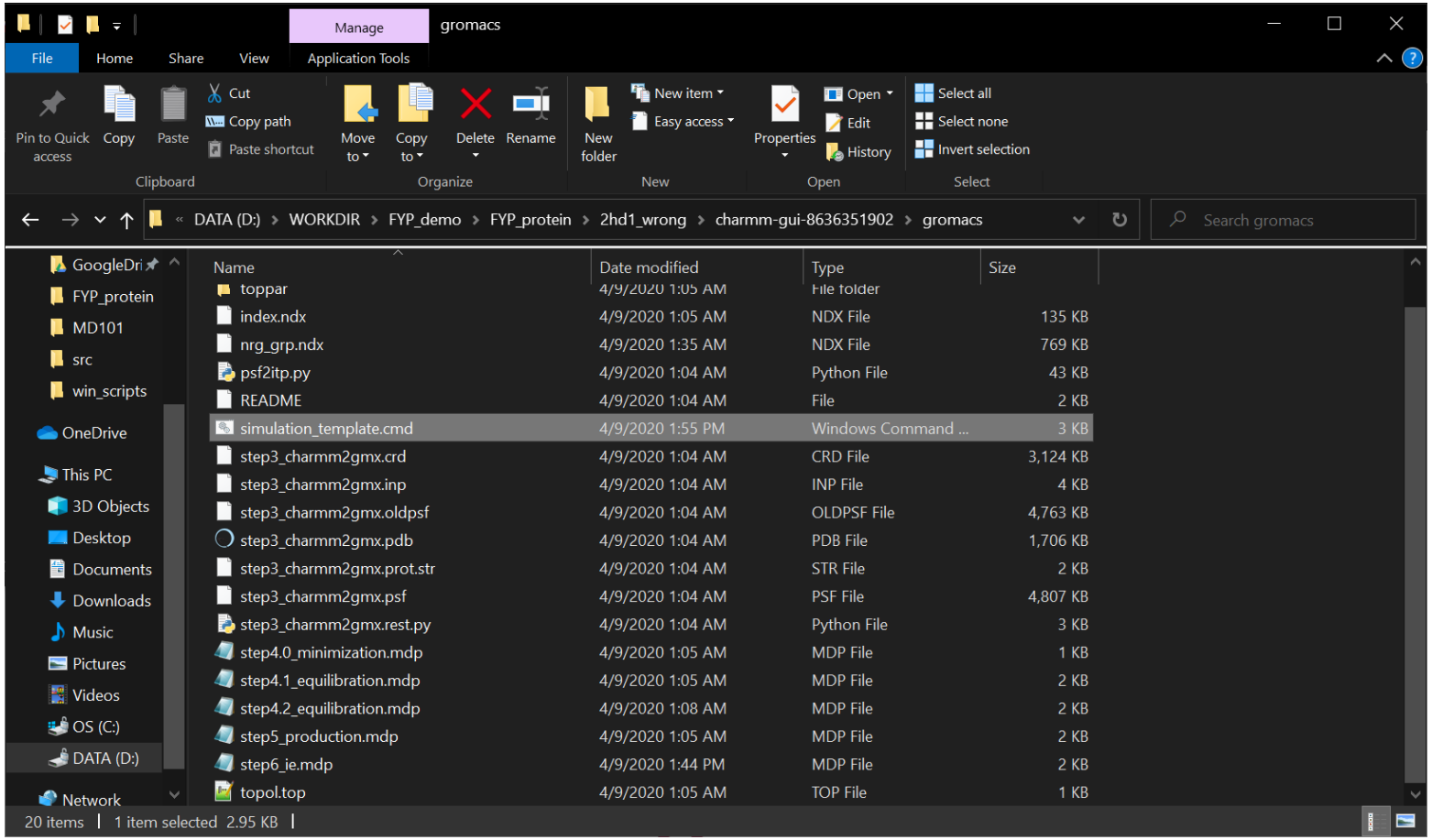
simulation_template.cmd

```
%GMXPATH%\gmx.exe grompp -f step4.2_equilibration.mdp -o step4.2_equilibration.tpr -c step4.1_equilibration.gro -r step3_charmm2gmx.pdb -n index.ndx -p
%GMXPATH%\gmx.exe mdrun -v -deffnm step4.2_equilibration
ECHO Density | %GMXPATH%\gmx.exe energy -f step4.2_equilibration.edr -o density.xvg
REM Production
%GMXPATH%\gmx.exe grompp -f step5_production.mdp -o step5_md.tpr -c step4.2_equilibration.gro -n index.ndx -p topol.top
%GMXPATH%\gmx.exe mdrun -v -deffnm step5_md
REM Post-process trajectory for visualization
ECHO 1 0 | %GMXPATH%\gmx.exe trjconv -s step5_md.tpr -f step5_md.xtc -o step5_md_noPBC0.xtc -pbc mol -center -ur compact
ECHO 1 0 | %GMXPATH%\gmx.exe trjconv -s step5_md.tpr -f step5_md_noPBC0.xtc -o step5_md_noPBC.xtc -fit rot+trans
REM RMSD analysis
ECHO 4 4 | %GMXPATH%\gmx.exe rms -s step5_md.tpr -f step5_md_noPBC.xtc -o rmsd.xvg -tu ns
ECHO 4 4 | %GMXPATH%\gmx.exe rms -s step4.0_minimization.tpr -f step5_md_noPBC.xtc -o rmsd_xtal.xvg -tu ns
%GMXPATH%\gmx.exe grompp -f step6_ie.mdp -c step4.2_equilibration.gro -p topol.top -n nrg_grp.ndx -o step6_ie.tpr
%GMXPATH%\gmx.exe mdrun -deffnm step6_ie -rerun step5_md.xtc -nb cpu
ECHO Coul-SR:RECEPTOR-LIGAND | %GMXPATH%\gmx.exe energy -f step6_ie.edr -o Coul-SR.xvg
ECHO LJ-SR:RECEPTOR-LIGAND | %GMXPATH%\gmx.exe energy -f step6_ie.edr -o LJ-SR.xvg
ECHO.
ECHO Your GROMACS job is finished!
PAUSE
```

Batch file | length : 3,080 lines : 51 | Ln : 48 Col : 83 Sel : 352 | 4 | Unix (LF) | UTF-8 | IN

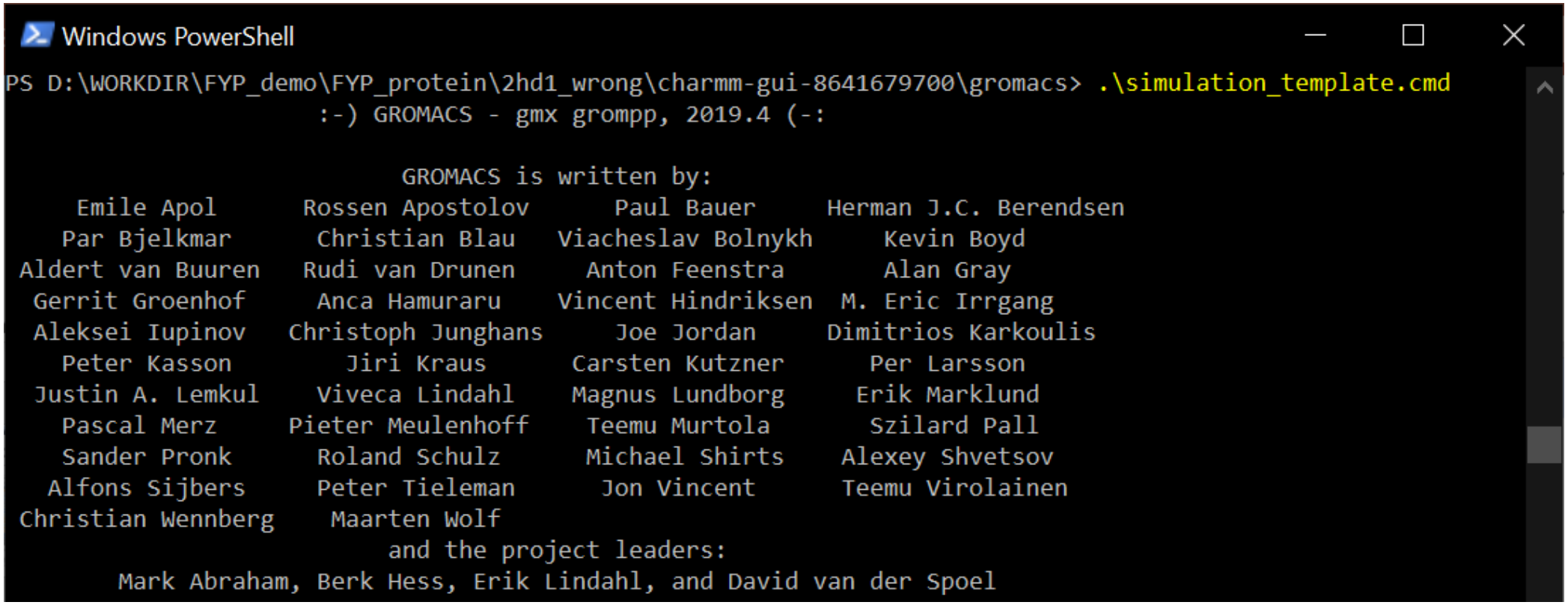
4. Check your input files again

- Check all the input files carefully before you start



5. Run the simulation

- This time, since we have some important information printed to standard output only, it is better to execute the script within the Windows PowerShell or command prompt.
- Remember to estimate the time required for the simulation!!!
  - Open this folder in the Windows PowerShell or command prompt.
  - Type: .\simulation_template.cmd



6. Analysis

- After the simulation ends (maybe half a day?):
  - Scroll up a bit, write down the average LJ-SR and Coul-SR energy with error estimation
  - Add them together to get the total interaction energy

Energy	Average	Err.Est.	RMSD	Tot-Drift	
LJ-SR:RECEPTOR-LIGAND	-104.088	2.4	7.69452	-7.60756	(kJ/mol)

Energy	Average	Err.Est.	RMSD	Tot-Drift	
Coul-SR:RECEPTOR-LIGAND	-2.07416	1.4	3.62329	-5.43909	(kJ/mol)

- **Note that this is NOT the free energy or binding energy that are generally used -** just the decomposed short-range interaction energy that may or may not be useful, depends on force fields.
  - Usually more complicated methods are used for a more accurate analysis.
  - However it is quick and maybe useful in your case.

6. Analysis

- Monitor the potential energy in EM, temperature and density in all the simulations…
  - Plot the corresponding .xvg files with QtGrace
- Visual the trajectory in e.g. Chimera, check if the broken chains moved too far away…
- Check the rmsd compared to crystal structure and throughout the simulation…
  - Plot rmsd_xtal.xvg and rmsd.xvg respectively